StatsBotEval: An Automated Evaluation Framework for Student-GenAI Interactions

Akshay Lakhi, Daniel Reiter, Deeviya Francis Xavier, Frank Scharnowski, Leonardo Bergmann
University of Vienna
lakhia92@univie.ac.at

1 Background

Usage of generative artificial intelligence (GenAI) by students can lead to performance gains for the learners, yet undermines the processes that are essential for durable learning (Yan et al., 2025). Further, increased dependence on GenAI seems to negatively affect problem-solving and critical thinking skills, unrestricted usage even impairing long-term knowledge retention. On the other hand, a recent meta-analysis on integrating the cognitive and emotional aspects of support in AI applications provided evidence for a strong effect on knowledge acquisition (H. Zhang et al., 2025). These contradictory findings underscore the tension in the usage of GenAI in educational contexts.

2 Research Proposal

Our project attempts to defuse this tension by systematically characterizing how students interact with GenAI, and how this information can inform an educator's teaching practice. We propose to develop *StatsBotEval*, an automated evaluation framework for student-GenAI interactions, where chats come from the *StatsBot,* a ChatGPT-like tool for learning statistics, in use by the psychology students at the University of Vienna since early 2025.

The research project consists of three milestones. First, we build an educator-facing dashboard that surfaces StatsBot conversational data in real time, displaying descriptive statistics on student-GenAI interactions including: (1) topic distribution, (2) temporal usage patterns, (3) deductive and inductive content classifications, (4) language patterns, and (5) usage context (e.g. number of users and questions). Within Bergmann et al.'s framework (2025), deductive classifications assign conversations to predefined, curriculum-derived topics (e.g., the statistics content listed in u:find), while inductive classifications derive finer, data-driven categories from the chat content itself. Second, we conduct an exploratory analysis of the data with machine learning techniques by relating student chat interactions to course performance and learning challenges. Finally, we consolidate the empirical findings into a master's thesis by situating them within the educational-GenAI literature.

3 Methods

We already have ethics approval for the project, and plan to develop the system with a privacy-first and GDPR-compliant approach: chats are linked to course records under pseudonymous identifiers with direct identifiers stripped before analysis, and the dashboard reports only aggregated, non-identifying outputs.

For the exploratory analysis, gradient boosted decision trees would be used to understand individual and collective student learning patterns, how these relate to course performance, and the challenges students face while learning. SHAP (SHapley Additive exPlanations) would then identify which learning behaviors drive these predictions, capturing not just each feature's magnitude but also its direction and interactions. We chose interpretable predictive ML over standard statistical methods to avoid certain known pitfalls, for instance, cross-validation gives out-of-sample prediction that standard inference usually does not.

4 Significance

StatsBotEval aims to serve as the evaluation framework for student-GenAI interactions. It does so by surfacing patterns and insights that instructors can act on to improve their teaching, and by characterizing the learning challenges students face. *StatsBotEval* would also help the University of Vienna make more informed decisions about how it integrates GenAI into its programs. Considering most educational GenAI research comes from East Asia and the US, this observational study adds a perspective from Central Europe, where students engage with these tools in German as well as English.

References

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